Retail Sales Analysis

Business Problem and Project Brief

Project Aim: Use retail transaction data to understand how sales, profit, order quantity, products, customers, locations, and time periods relate, then present the results in an interactive Power BI dashboard to support better business decisions.

Business problem

Retail managers need a clear view of what drives revenue and profit across products, customer segments, regions, and time. Without consistent preparation and analysis, transaction records make it difficult to spot underperforming areas, compare performance fairly, or identify changes in demand.

This project asks: How can retail sales data reveal performance trends and areas for investigation so the business can improve profitability and make better product, customer, and regional decisions?

Project objectives

- Clean and prepare the data in Python by checking structure, data types, missing values, duplicates, and consistency.
- Store the prepared dataset in a MySQL database managed through XAMPP for structured access and querying.
- Build an interactive Power BI dashboard for sales, profit, quantity, and trends across key business dimensions.
- Turn verified patterns into clear findings and practical recommendations.

Data and analysis scope

The dataset contains 10,194 sales-line records with order and shipping dates, customer and product fields, segment, region and state, category and sub-category, sales, quantity, discount, and profit. The analysis describes patterns in these records; it does not prove why they occurred.

Business questions

- What are the total sales, total profit, and total order quantity for the selected year?
- How does total order quantity change month by month across the selected years?
- How do monthly sales trends differ among Furniture, Office Supplies, and Technology?
- How does profit margin differ among Consumer, Corporate, and Home Office segments by year?
- How do monthly sales change during the year, and which months show the highest or lowest sales?
- How is total profit distributed among the West, East, South, and Central regions?
- How does total profit vary by quarter and year, and which quarters show the highest or lowest profit?

Project approach

1. **Prepare.** Use Python to inspect and clean the source data, preserve valid sales lines, standardize types, and document corrections.
2. **Store.** Load the prepared data into MySQL in XAMPP and confirm the schema, row count, and data types.
3. **Model.** Import the data into Power BI, set date and numeric types, create a date table, and define reusable measures.
4. **Explore.** Compare periods, products, regions, segments, discounts, and shipping using consistent filters.
5. **Communicate.** State each verified finding with its time period and filters, then connect recommendations to the evidence.